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They do not affect the animal's susceptibility to parasites.

•

They increase the chances that the animal will be selected as a mate.

Inference

The passage states that carotenoids are used by the immune system and for detoxification processes. To the extent that any carotenoids are used for these purposes, the passage suggests, they would not also be available for bright-colored display to potential mates.

A. The passage states that animals can acquire carotenoids from insects, as well as from plants and algae. Nothing indicates that a male animal's use of the carotenoids would determine which of these sources they are acquired from.

B. No information in the passage suggests this. The passage tells us that males acquire carotenoids by foraging.

C. **Correct.** The passage implies that the quantity of carotenoids used for detoxification is no longer available for any other purpose, such as display of bright coloration.

D. The passage suggests that carotenoids could benefit immune response; this implies the possibility of better resistance to parasitic infections.

E. The passage conjectures that a male animal's use of carotenoids for detoxification processes could reduce the chances of being selected as a mate.

The correct answer is C.

Questions 527–531 refer to the passage on page 462.

Main Idea Summary

The passage is a discussion of historian Linda Kerber's "republican motherhood" thesis, according to which the American Revolution led to an increase in educational opportunities for women and girls in the United States in the late eighteenth century. Kerber's work was influential at least partly because she explored a topic (female education) that historians had largely ignored. However, the passage cites an earlier historian who had provided evidence that the trend toward greater educational opportunities for women and girls had begun decades before the American Revolution. The main idea of the passage is that the influence of the "republican motherhood" thesis may have led historians to overlook evidence that the trend Kerber describes was already underway before the American Revolution.

527. According to the passage, Kerber maintained that which of the following led to an increase in educational opportunities for women in the United States after the American Revolution?

A. An unprecedented demand by women for greater educational opportunities in the decades following the Revolution

B. A new political ideology calling for equality of opportunity between women and men in all aspects of life

C. A belief that the American educational system could be reformed only if women participated more fully in that system

D. A belief that women needed to be educated if they were to contribute to the success of the nation's new form of government

E. A recognition that women needed to be educated if they were to take an active role in the nation's schools and churches

Supporting Idea

The passage ascribes to Linda Kerber the claim that there was *a surge of educational opportunities for women in the United States* after the American Revolution, and that this surge resulted from a new ideology of "republican motherhood." According to the passage, Kerber argued that the nation's leaders advocated education for women to equip them, in their family role, to raise politically virtuous sons.

A. The passage attributes no claim to Kerber concerning a demand by women for education.

B. The passage attributes no claim to Kerber concerning a new ideology calling for equality between women and men.

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and men.

C. Kerber's argument as represented in the passage did not claim that an increase in education opportunities for women resulted from a belief that such an increase was required for successful reform of the American educational system.

D. **Correct.** According to the passage, Kerber argued that educational opportunities for women increased because the nation's leaders believed that successful democratic government would require that women raise politically virtuous sons within their families, and that women could do so only if they had access to education themselves.

E. According to the passage, Kerber's thesis primarily concerns the roles that it was believed educated women could play in raising politically virtuous sons in the context of the family, not in the nation's schools or churches.

The correct answer is D.

528. According to the passage, within the field of educational history, Thomas Woody's 1929 work was

A. innovative because it relied on newspaper advertisements as evidence

B. exceptional in that it concentrated on the period before the American Revolution

C. unusual in that it focused on educational attitudes rather than on educational practices

D. controversial in its claims regarding educational opportunities for boys

E. atypical in that it examined the education of girls

Supporting Idea

According to the passage, Woody's work was a *notable exception* as contrast to the work of other educational historians, who *barely mentioned women and girls*.

A. Other historians prior to Woody's 1929 work may have used newspaper advertisements as evidence, but the passage provides no information as to whether this was so.

B. The passage is silent as to whether educational historians besides Woody concentrated on the period before the American Revolution.

C. The passage does not provide information as to the extent to which either Woody or other historians focused on educational attitudes as opposed to educational practices.

D. According to the passage, Woody noted that educational opportunities increased for both girls and boys around 1750. But the passage does not indicate that this claim, or any other claim Woody may have made about educational opportunities for boys, was controversial.

E. **Correct.** As stated above, the passage describes Woody's work as a *notable exception*, i.e., atypical, with respect to his discussion of education for girls.

The correct answer is E.

529. The passage suggests that Woody would have agreed with which of the following claims regarding "An Essay on Woman"?

A. It expressed attitudes concerning women's education that were reflected in new educational opportunities for women after 1750.

B. It persuaded educators to offer greater educational opportunities to women in the 1750s.

C. It articulated ideas about women's education that would not be realized until after the American Revolution.

D. It offered one of the most original arguments in favor of women's education in the United States in the eighteenth century.

E. It presented views about women's education that were still controversial in Woody's own time.

Application

According to the passage, Woody characterized "An Essay on Woman" (1753) as reflecting a shift in view, and the context indicates that this shift concerned new attitudes that accompanied increased opportunities after 1750 for girls to become educated women.

A. **Correct.** Based on the passage, this is a claim with which Woody would likely have agreed.

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A. **Correct.** Based on the passage, this is a claim with which Woody would likely have agreed.

B. The passage represents Woody as claiming that "An Essay on Woman" reflected changes that had already occurred around 1750. The passage does not indicate whether Woody would have agreed with this claim about a persuasive effect on educators.

C. Nothing in the passage represents Woody as thinking that "An Essay on Woman" had ideas about women's education that did not come to fruition until after the American Revolution. The tenor of Woody's thinking, as the passage represents it, is that the essay reflected changes already occurring.

D. The passage indicates that Woody characterizes "An Essay on Woman" as *reflecting* a view that had already gained some currency, so it is unlikely that Woody saw the essay as offering any highly original arguments in favor of women's education.

E. It may be true that "An Essay on Woman" presented some views that were at least somewhat controversial even around 1929, but the passage provides no information that addresses this point.

The correct answer is A.

530. The passage suggests that, with regard to the history of women's education in the United States, Kerber's work differs from Woody's primarily concerning which of the following?

A. The extent to which women were interested in pursuing educational opportunities in the eighteenth century

B. The extent of the support for educational opportunities for girls prior to the American Revolution

C. The extent of public resistance to educational opportunities for women after the American Revolution

D. Whether attitudes toward women's educational opportunities changed during the eighteenth century

E. Whether women needed to be educated in order to contribute to the success of a republican form of government

Evaluation

The passage represents Kerber as claiming that the American Revolution led to a surge in educational opportunities for women because the nation's leaders believed women needed to be educated if they were to raise politically virtuous sons. Woody, however, is represented as claiming that there was a significant increase in such opportunities and significant advocacy for women's education well before the Revolution.

A. The passage does not represent either Kerber or Woody as addressing the extent to which women were interested in pursuing educational opportunities in the eighteenth century.

B. **Correct.** The passage attributes to Woody the view that *practical education for females had many advocates before the Revolution*, notably in the 1750s, and that the Revolution at most accelerated an earlier trend of changing attitudes. This is contrary to the views attributed to Kerber.

C. The passage gives no information as to whether Kerber or Woody addresses this issue, nor does it discuss to what extent, if any, such resistance may have occurred.

D. The passage indicates that Kerber and Woody hold that there was a change in attitudes toward women's educational opportunities during the eighteenth century, disagreeing, however, as to whether the most significant change occurred before or after the Revolution.

E. Neither Kerber nor Woody is represented by the passage as holding divergent views on this point, and it would be reasonable to think that they may have agreed.

The correct answer is B.

531. According to the passage, Kerber argued that political leaders thought that the form of government adopted by the United States after the American Revolution depended on which of the following for its success?

A. Women assuming the sole responsibility for instilling political virtue in children

B. Girls becoming the primary focus of a reformed educational system that emphasized political virtue

C. The family serving as one of the primary means by which children were imbued with political virtue

D. The family assuming many of the functions previously performed by schools and churches

E. Men and women assuming equal responsibility for the management of schools, churches, and the family

Supporting Idea

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The passage attributes to Kerber the claim that the nation's leaders believed a virtuous citizenry was essential to the success of the nation's republican form of government, and that women would play a primary role in raising future citizens who would be politically virtuous.

A. According to the passage, Kerber indicates that the nation's leaders believed churches and schools, as well as families, would work to imbue political virtue, though they emphasized the crucial role of families.

B. Kerber argues that the educational system underwent reform in the sense that educational opportunities for women increased but does not claim that schools or families would change focus to imbue girls with political virtue.

C. **Correct.** Kerber argues that political leaders emphasized the family as the primary means by which future citizens would be imbued with political virtue.

D. Kerber does not claim the nation's leaders proposed that the family would take over functions previously fulfilled by schools and churches.

E. Kerber does not attribute to the nation's leaders the view that men and women would exercise equal roles in managing schools, churches, and the family.

The correct answer is C.

Questions 532–536 refer to the passage on page 464.

532. It can be inferred from the passage that geologists today would be most likely to agree with which of the following statements about Wegener's *The Origin of Continents and Oceans*?

A. It was a worthy scientific effort that was ahead of its time.

B. It was based on evidence that was later disproved.

C. It was directly responsible for the acceptance of the theory of plate tectonics.

D. It has been disproved by continental drift theory.

E. It misrepresented how horizontal displacements cause the formation of mountain chains.

Inference

Notice what is asserted in the passage's first paragraph about the attitudes or beliefs of today's geologists. We read that most geologists today consider Wegener's book "an impressive and prescient document" because its ideas significantly overlap with those of plate tectonics, the theory accepted by most geologists today. But the passage stops short of attributing the development of plate tectonics to Wegener's ideas, which were widely rejected by Wegener's peer geologists.

A. **Correct.** The passage tells us that today's geologists find Wegener's book to be impressive, i.e., respectable as a work of science, as well as "prescient," i.e., it anticipated some of the key assumptions of plate tectonics.

B. Nowhere in the passage is it asserted that Wegener's evidence was "disproved," though the passage indicates that Wegener's evidence and his theoretical findings were largely rejected by his geological contemporaries. The passage does not suggest that the rejection was based on countervailing evidence.

C. The word "directly" in this option is unwarranted by any information provided in the passage, even though the passage notes an important conceptual kinship between Wegener's theories and plate tectonics.

D. The passage tells us that the content of Wegener's book focused on what was later called continental drift theory. Rather than supporting the claim that continental drift theory disproved the major ideas in Wegener's book, it shows the absurdity of such a claim.

E. This option suggests that today's geologists assume Wegener's theory lacked an adequate account of mountain range formation. The passage suggests that today's geologists believe Wegener's theory lacked "an adequate mechanical basis." In other words, it provided no plausible mechanism to adequately explain how continental drift produced mountain ranges. This is distinct from misrepresenting the process, i.e., providing an account of the process that was false or misleading.

The correct answer is A.

533. The author of the passage suggests the most likely explanation for the geological community's response to continental drift theory in its day was that the theory

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A. was in conflict with certain aspects of plate tectonics theory

B. failed to account for how mountains were formed

C. did not adequately explain how continents moved through the ocean floor

D. was contradicted by the geophysical data of the time

E. was based on a kind of evidence that was considered insufficiently convincing

Supporting Idea

Why did Wegener's scientific peers reject his continental drift theory? In the passage's second paragraph, the author suggests that the most likely cause was "the nature of the evidence" Wegener offered in support of it. Wegener's evidence was derived from "direct observations of rocks in the field" using simple tools such as hammers and hand lenses, whereas plate tectonics used technically sophisticated instruments.

A. The passage indicates that the primary motivation for the rejection of continental drift theory by Wegener's peers was not theoretical incompatibility with plate tectonic theory. The passage emphasizes the significant overlaps in the "presumptions" of both theories.

B. The passage states that Wegener's theory was rejected despite providing a possible explanation for mountain formation, not because it failed to do so.

C. The author of the passage signals disagreement with the view of a number of other researchers that Wegener's peers rejected continental drift theory because it failed to explain how continents could move through the ocean floor.

D. The passage neither states nor implies that continental drift theory conflicted, or was perceived to conflict, with geophysical data available around 1912 when Wegener's book was published.

E. **Correct.** The passage contrasts the kind of data related to "similarities of patterns and forms" with the more precise kind of data that plate tectonics collected. Wegener's data was collected using hammers and hand lenses, whereas the data collected by plate tectonics used technical instruments. This difference is offered to explain why Wegener's peers found his data less convincing. According to the author, this likely led to their rejection of continental drift theory.

The correct answer is E.

534. The author of the passage probably refers to the "considerable overlap" (See line 18) between the continental drift theory and plate tectonics theory in order to

A. suggest that plate tectonics theory is derived from Wegener's work

B. introduce a discussion comparing the elements of the two theories

C. examine the question of whether continental drift theory was innovative in its time

D. provide a reason why it might seem surprising that continental drift theory was not more widely embraced by geologists

E. cite an explanation that has been frequently offered for Wegener's high standing among geologists today

Supporting Idea

The sentence in line 18 has the following structure: Despite X and despite Y, what occurred was Z. The point being made is that it was surprising that Z occurred when X and Y would have led us to expect that Z would not occur. More concretely: Despite continental drift's overlap with plate tectonics and its initially plausible explanation for the origin of mountains, it was surprising that Wegener's peer geologists largely rejected continental drift theory.

A. The author does not imply anywhere that the plate tectonics were derived from continental drift theory. In fact, the passage describes a difference in the kind of data and collection methods used by the two theories.

B. No such discussion occurs following the phrase in line 18.

C. The passage suggests that continental drift theory was innovative in 1912, but there is no discussion in the context of line 18 as to whether, or in what respects, it was innovative.

D. **Correct.** See the introductory analysis above. The sentence in line 18 provides two reasons why one would expect that continental drift theory would have gained acceptance among Wegener's peer geologists. Surprisingly, it did not.

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Surprisingly, it did not.

E. The context of line 18 focuses on explaining the surprising nonacceptance of Wegener's continental drift theory by his contemporaries in geology. That context does not address Wegener's standing among today's geologists.

The correct answer is D.

535. The author cites Hallam (see line 32) on the ice ages primarily in order to

A. provide an example of a geologic phenomenon with precise causes that are not fully understood by geologists today

B. criticize the geological community for an apparent lack of consistency in its responses to new theories

C. offer evidence held to undermine a common view of why Wegener's theory was not accepted in its day

D. give an example of a modern scientist who believes Wegener's theory was rejected because it failed to adequately explain the mechanical basis of continental drift

E. support some researchers' rationale for why Wegener's theory was rejected by most geologists in the early twentieth century

Evaluation

The point attributed to Hallam is that the acceptance of a new scientific phenomenon by the scientific community has often not required that the phenomenon have a satisfactory explanation provided for it. What is the author's purpose in citing this point? The point immediately preceding this one is that most of today's geologists and historians of science believe that Wegener's continental drift theory was rejected because it was deficient in explaining some of the phenomena it cited. The author disagrees with this view and cites Hallam as evidence to counter this widely held view of why Wegener's theory was not accepted by his contemporaries in geology.

A. The passage does not provide any example of a phenomenon, the occurrence of which Wegener established and today's geologists regard as not yet having received an adequate causal explanation.

B. The passage asserts that geologists and historians today believe that what caused the nonacceptance of Wegener's theory after 1912 was a deficiency in its explanatory power. But the point attributed to Hallam suggests that this historical account is inadequate. The author cites Hallam not primarily to "criticize" the geological community for a general lack of consistency in "responses to new theories," but to help show the inadequacy of a widely accepted historical explanation.

C. **Correct.** The purpose of the citation of Hallam is to help show the inadequacy of a widely accepted historical explanation concerning the nonacceptance of Wegener's theory by his contemporaries.

D. The citation of Hallam, a twentieth-century geologist, indicates that he probably did not endorse the historical explanation, which is widely accepted by most of today's geologists and science historians, of why Wegener's theory was not accepted.

E. The citation of Hallam is intended to cast doubt on the historical explanation offered by other researchers, not to support it.

The correct answer is C.

536. The author of the passage most likely discusses the "essential presumptions" (see lines 12–13) of *The Origin of Continents and Oceans* in order to

A. indicate features of Wegener's theory that caused it to be doubted in its day

B. show why Wegener's theory is now regarded as prescient

C. indicate differences between plate tectonics and the theory of continental drift

D. cite features of the theory of continental drift for which no evidence was available in Wegener's day

E. point out aspects of Wegener's theory that were accepted well before the advent of plate tectonics

Evaluation

According to the passage, Wegener's book is regarded as "prescient" by most geologists today. This point is supported by detailing three "essential presumptions" of Wegener's theory that are shared by plate tectonics. Thus, the author aims to show that most geologists see Wegener's theory as anticipating, in some important

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respects, the scientific advances achieved by plate tectonics.

A. The passage does not suggest that the presumptions that Wegener's theory shared with plate tectonics were the cause of the nonacceptance of the theory by Wegener's contemporary peers in geology.

B. **Correct.** The context of the phrase "essential presumptions" indicates that the author's purpose is to show how Wegener anticipated some of the important ideas in plate tectonics and was therefore prescient.

C. The passage indicates that there were such differences, but that is not the author's goal in mentioning the "essential presumptions" shared by plate tectonics and continental drift theory.

D. The passage suggests that the kind of evidence given by Wegener for continental drift theory differed from that on which plate tectonics relied, but the author's purpose in citing the "essential presumptions" is not to indicate deficiencies in the evidence for claims made by continental drift theory.

E. The passage does not tell us whether the "essential presumptions" cited were or were not accepted by Wegener's contemporaries in geology.

The correct answer is B.

Questions 537–541 refer to the passage on page 466.

537. The passage suggests that in order for a manufacturer in a capital-intensive industry to have a decisive advantage over competitors making similar products, the manufacturer must

A. be the first in the industry to build production facilities of theoretically optimal size

B. make every effort to keep fixed and sunk costs as low as possible

C. be one of the first to operate its manufacturing plants at minimum efficient scale

D. produce goods of higher quality than those produced by direct competitors

E. stockpile raw materials at production sites in order to ensure a steady flow of such materials

Inference

This question asks for an inference about what a manufacturer in a capital-intensive industry must do to have an advantage over competitors making similar products. The passage addresses this question by stating that advantage accrues to those firms that are the first to exploit the full potential of optimally sized, technologically sophisticated plants. In this context, exploiting the full potential of such plants means operating them at *minimum efficient scale*. Based on the definition in the first paragraph, this means that the plant must have an output of such a size that the cost per unit of output is at a minimum.

A. The passage says that for new capital-intensive firms to dominate the market, it is not enough for them to have optimally sized plants; the plants must also be operated in a way that fully exploits their potential.

B. While keeping fixed and sunk costs low would obviously help keep overall costs low, the passage does not suggest that this is decisive in enabling a firm to have an advantage over competitors.

C. **Correct.** Being among the first manufacturers to operate plants at minimum efficient scale means that those plants are being exploited to their full potential. This strategy would most likely give such manufacturers a decisive advantage over new firms hoping to compete effectively.

D. The passage does not discuss the quality of goods made by manufacturers.

E. The passage does not suggest that stockpiling raw materials is the most efficient way to ensure a steady flow of raw materials into the manufacturing process, though the passage states that such a steady flow is a factor in achieving minimum efficient scale.

The correct answer is C.

538. The passage suggests that which of the following is true of a manufacturer's fixed and sunk costs?

A. The extent to which they are determined by market conditions for the goods being manufactured is frequently underestimated.

B. If they are kept as low as possible, the manufacturer is very likely to realize significant profits.

C. They are the primary factor that determines whether a manufacturer will realize economies of scale.

D. They should be on a par with the fixed and sunk costs of the manufacturer's competitors.

E. They are not affected by fluctuations in a manufacturing plant's throughput.

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E. They are not affected by fluctuations in a manufacturing plant's throughput.

Inference

This question asks about what the passage implies about fixed and sunk costs. The passage states that when production declines due to certain factors, such costs remain at the same level (which may be high), and the cost per unit produced (*unit costs*) rises sharply.

A. The passage discusses the impact of market conditions on determining what the optimal size of a manufacturing plant is (which affects fixed and sunk costs). But it makes no claim about the frequency with which such an impact is "underestimated."

B. The passage emphasizes that failing to keep throughput at an efficiently high level reduces profitability because that failure results in increased cost per unit (to which, of course, the plant's fixed and sunk costs contribute). But the passage does not claim that keeping aggregate fixed and sunk costs very low is necessary in order to have the most competitive production operation.

C. The passage emphasizes that the crucial factor in achieving economies of scale is efficient operation of the production facilities, not the size of the firm's fixed and sunk costs (even though such costs are clearly in part determined by the size and design of the production facilities).

D. While a manufacturer's fixed and sunk costs may be on a par with those of the manufacturer's competitors, the passage provides no grounds for inferring that there is any need for them to be (for example, physical plants that employ different technologies may have different price tags).

E. **Correct.** According to the passage, *throughput* refers to the flow of materials through a plant. This flow can vary as a result of various factors, but fixed and sunk costs—financial resources already committed—remain the same regardless of such variation.

The correct answer is E.

539. In the context of the passage as a whole, the second paragraph serves primarily to

A. provide an example to support the argument presented in the first paragraph

B. evaluate various strategies discussed in the first paragraph

C. introduce evidence that undermines the argument presented in the first paragraph

D. anticipate possible objections to the argument presented in the first paragraph

E. demonstrate the potential dangers of a commonly used strategy

Evaluation

This question asks about the rhetorical function of the second paragraph. While the first paragraph argues that a crucial factor in achieving economies of scale is intangible capital, or organized human capabilities, the second paragraph uses the example of new capital-intensive manufacturing industries to help show that this is indeed the case.

A. **Correct.** The second paragraph provides an example that illustrates the claims made in the first paragraph. It discusses the way in which intangible capital—e.g., distribution networks, marketing systems, smooth production processes, and qualified management teams—enables manufacturers in new capital-intensive manufacturing industries to realize economies of scale and achieve market dominance.

B. The second paragraph does, in a sense, "evaluate" investment in intangible capital: it suggests that such investment is necessary. However, investment in intangible capital is the only strategy it discusses.

C. The second paragraph supports rather than undermines the first paragraph's argument.

D. Nothing in the second paragraph suggests that there are, or could be, any objections to the first paragraph's argument.

E. The second paragraph discusses the potential positive outcomes of investing in intangible capital. It suggests that there might be negative consequences to not making such investments, but it does not indicate that avoiding such investments is a commonly used strategy.

The correct answer is A.

540. The passage LEAST supports the inference that a manufacturer's throughput could be adversely affected by

A. a mistake in judgment regarding the selection of a wholesaler

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A. a mistake in judgment regarding the selection of a wholesaler

B. a breakdown in the factory's machinery

C. a labor dispute on the factory floor

D. an increase in the cost per unit of output

E. a drop in the efficiency of the sales network

Application

This question may be best approached by using an elimination strategy—first finding the four choices that can reasonably be inferred from the passage, and then checking to make sure that the remaining choice cannot reasonably be inferred. This requires understanding the information the passage gives about throughput, then making inferences about what can cause throughput to drop. The passage defines throughput generally as *the flow of materials through a plant* and goes on to explain that it involves coordination of the production process itself, as well as obtaining materials from suppliers and marketing and distributing the manufactured products. Anything that damages this flow of materials and products would be said to have an adverse effect on throughput.

A. Making a poor judgment about a wholesaler would most likely have an adverse effect on throughput, in that it could affect *the flow of output to wholesalers and final consumers*.

B. A breakdown in machinery would likely fall into the category of *problems on the factory floor* mentioned in the passage and would likely prove damaging to throughput because of its effect on the production process itself.

C. A labor dispute would also likely fall into the category of *problems on the factory floor* mentioned in the passage and would probably cause a decline in production and thus adversely affect throughput.

D. **Correct.** The passage emphasizes that changes in throughput can cause increases or decreases in costs per unit. But the passage is not committed to any claims about how changes in costs per unit might affect throughput.

E. The passage suggests that inefficient sales networks could cause a decline in production. Thus, a decrease in sales efficiency would most likely adversely affect a manufacturer's ability to provide goods to consumers, and therefore would create problems with throughput.

The correct answer is D.

541. The primary purpose of the passage is to

A. point out the importance of intangible capital for realizing economies of scale in manufacturing

B. show that manufacturers frequently gain a competitive advantage from investment in large manufacturing facilities

C. argue that large manufacturing facilities often fail because of inadequate investment in both tangible and intangible capital

D. suggest that most new industries are likely to be dominated by firms that build large manufacturing plants early

E. explain why large manufacturing plants usually do not help manufacturers achieve economies of scale

Main Idea

This question depends on understanding the passage as a whole. In general, it makes an argument for investing in intangible capital as a way for manufacturers to realize economies of scale, and it supports its argument with an example.

A. **Correct.** The passage focuses on intangible capital as a crucial factor in realizing economies of scale.

B. According to the passage, manufacturers gain competitive advantage by building plants of optimal size that they then fully exploit; nothing in the passage suggests that large plants are frequently optimal.

C. The passage assumes that manufacturers invest appropriately in tangible capital and argues that it is important for them to invest in intangible capital as well.

D. The passage states that new capital-intensive manufacturing industries are dominated not by firms that are the first to build large plants, but by firms that exploit the full potential of their plants.

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E. The passage indicates that economies of scale can be achieved in plants of optimal size. The passage does not suggest that large plants cannot be optimal.

The correct answer is A.

Questions 542–545 refer to the passage on page 468.

Main Idea Summary

The primary purpose of the passage is to describe resin and discuss a theory on its biological function. The first paragraph identifies some of the characteristics of resin and distinguishes it from gum and sap; one way in which resin differs from these two substances is that its physiological function is not clearly understood. The second paragraph considers whether resin may function to repel or attract insects, and indicates that the consideration of such issues led to the development of the discipline of chemical ecology.

542. According to the passage, which of the following is true of plant antiherbivore chemistry?

A. Changes in a plant's antiherbivore chemistry may affect insect feeding behavior.

B. A plant's repellent effects often involve interactions between gum and resin.

C. A plant's antiherbivore responses assist in combating bacterial infections.

D. Plant antiherbivore chemistry plays only a minor role in the coevolution of plants and insects.

E. Researchers first studied repellent effects in plants beginning in the 1950s.

Supporting Idea

This question addresses what the information in the passage indicates about plant antiherbivore chemistry—that is, plants' chemical defenses against herbivore attacks. The second paragraph of the passage cites the views of various scientists regarding the possible role of resin in antiherbivore chemistry; plants could have evolved resin specifically to repel insects.

A. **Correct.** According to the second paragraph, various scientists have suggested that a change in antiherbivore chemistry, here

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